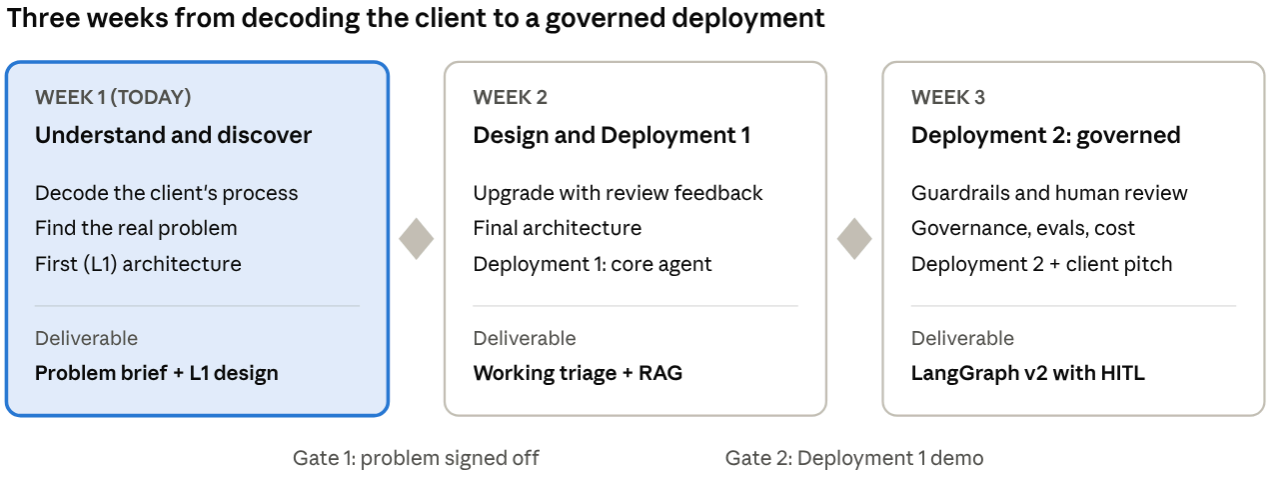
Session 1 · Decoding the client

Today we meet the client. We will learn who Ramco Payce is, how payroll works in India and the UAE, how their customers raise support tickets, and what each stakeholder told us. We won't design anything today. The goal is to understand the problem well enough to know what *not* to build.

> **Framing for the class:** Ramco Payce is a real company, and we will work as if it were our client. The support data, people and interviews in our data room are a realistic simulation built for this course; they are not Ramco's data.

Overview: the 3-week project



the 3-week project · 6 hours a week

Week	Focus	We finish the week with
1	Understand the client's process, discover the problem, first (L1) architecture	A two-page problem brief with evidence, CFO-ready metrics, and an L1 architecture
2	Upgrade the architecture with review feedback, lock the final architecture, Deployment 1	A working first deployment: intake, triage, routing and dated retrieval
3	Deployment 2 with governance and guardrails	A LangGraph agent with human approval, guardrails, caching, evaluations and a cost model, pitched to the client

> **Say it like this:** "An FDE's first deliverable is never code. It is a sentence the client agrees with: 'This is the problem, and this is how we will know it's fixed.' That is what Week 1 produces."

Who the client is

Our client is **Ramco Payce**, the payroll platform from Ramco Systems. The company is real. The support tickets, people and interviews we work with are a realistic simulation built for this course; they are not Ramco's data or staff.

What	Public fact	Source
Parent company	Ramco Systems; Payce sits in its Global Payroll business	ramco.com
Product	Ramco Payce, a platform-based payroll software, launched 9 Feb 2024	Ramco press release
Reach	Payroll coverage in 150+ countries; 500+ global payroll customers (company figures)	Ramco press release
Middle East	Launched in the UAE in Dubai on 7 May 2024, with a Riyadh launch planned	Gulf News
Features it advertises	No-code rule builder, country-specific prebuilt configurations, an anomaly and reasoning engine for payroll variances, a payroll chatbot, self-service reporting	Gulf News, press release

> **Say it like this:** "Ramco already sells a payroll chatbot. So when the CEO in our case says 'just build a chatbot', ask why the one they have isn't fixing support. That question is where discovery starts."

Our simulated support organisation: 42 agents in three tiers (L1, L2, L3) across Pune, Bengaluru and Dubai, serving India and UAE customers through email, an in-app widget, a partner portal, WhatsApp and phone. Enterprise contracts pay SLA credits when a deadline is missed. Full detail is in the Company brief and Support operation & data tabs.

> **Ask the class:** "What does a payroll company actually sell?" *Expected answer:* not software, but payday. Every employee paid correctly and on time, and every statutory filing right. Support tickets are about that promise breaking.

Payroll domain: India vs UAE

Payroll is gross pay minus deductions, paid on time, with the right files sent to the right authority. In India most of the difficulty is the **deductions**: four statutory schemes, each with its own rules and dates. In the UAE there is no income tax, so most of the difficulty is the **payment itself**: salaries must go through the Wage Protection System on time and in the right file format.

> **In plain words:** in India the support question is usually "why is this number different?" In the UAE it is usually "why did the bank reject my salary file?"

Topic	India	UAE
Income tax	**TDS**: the employer deducts income tax monthly using the employee's chosen regime and slabs; Form 16 at year end	No personal income tax on salary
Retirement / social security	**PF (EPF)**: 12% employee + 12% employer on wages up to a ceiling; the employer share is split between EPF and the pension scheme (EPS)	Expats: **end-of-service gratuity** paid when employment ends. UAE nationals: pension and social security contributions instead
Health insurance	**ESI**: employer + employee contributions for lower-wage employees, below a wage limit	Employer-provided health insurance (no payroll deduction scheme like ESI)
State-level tax	**Professional Tax (PT)**: a small monthly tax with slabs that differ by state	None
How salaries are paid	Bank transfer (NEFT/RTGS bulk files)	**WPS**: salaries go through banks or approved exchanges using a SIF (Salary Information File); the ministry tracks every payment
Payment deadline	Set by contract and practice; statutory deposits have their own due dates (PF/ESI by the 15th of the next month)	Since 1 Jun 2026 (Resolution 0340 of 2026), wages are due by the 1st of the next month; at least 85% must be paid. Enforcement steps start on day 2 (alerts), then day 5 (work-permit suspension), escalating to legal action by day 21

Topic	India	UAE
Monthly filings	PF ECR file, ESI return, TDS returns, PT returns	WPS SIF file each pay cycle
Year-end and exit	Form 16, full and final settlement, gratuity under Indian law after 5 years	Gratuity: no gratuity under 1 year; 21 days' basic salary per year for the first 5 years, 30 days per year after that; calculated on the last basic salary
Working week	Mostly Mon–Sat or Mon–Fri	Mon–Fri; time-zone gap of 1.5 h with India
Data protection	**DPDP Act 2023**; the DPDP Rules were notified in Nov 2025 and phase in through May 2027	**PDPL**, Federal Decree-Law No. 45 of 2021, in force since 2 Jan 2022
Languages in tickets	English, Hindi, Hinglish	English, Arabic, and many expat languages

Worked example: one month's payslip in each country

An employee in Mumbai earns ₹1,50,000 a month (basic ₹75,000) and picks the new tax regime. A colleague in Dubai earns AED 20,000 a month (basic AED 10,000). Rates are FY 2026-27 rules as published; the numbers are rounded and illustrative.

Line	India (₹/month)	UAE (AED/month)
Gross	1,50,000	20,000
Income tax	−12,567 (TDS, see below)	0 (no income tax)
PF	−1,800 (12% × ₹15,000 cap)	0 (no PF for expats)
ESI	0 (only if gross ⩽ ₹21,000)	0 (employer insurance)
Prof. Tax	−200 (Maharashtra)	0
Take-home	**1,35,433** via NEFT file	**20,000** via WPS SIF
Employer extra	PF 1,800 (EPS 1,250 + EPF 550)	Gratuity builds up

TDS, worked out: yearly 18,00,000 − 75,000 standard deduction = 17,25,000 taxable. Slabs: 0–4L nil, 4–8L 5% = 20,000, 8–12L 10% = 40,000, 12–16L 15% = 60,000, 16–17.25L 20% = 25,000. Total 1,45,000 + 4% cess = 1,50,800 a year, or 12,567 a month.

ESI, if the salary were lower: at ₹20,000 gross, the employee pays 0.75% = ₹150 and the employer 3.25% = ₹650. Maharashtra PT is ₹300 in February.

Gratuity in the UAE, worked out: the Dubai employee leaves after 7 years. That's 5 years × 21 days + 2 years × 30 days = 165 days of basic salary. With a daily basic of AED 10,000 ÷ 30 = AED 333.33, the gratuity is about **AED 55,000** (some employers divide by 365 instead, which gives slightly less). UAE nationals get pension contributions instead of gratuity, so this example covers expats only.

Where support tickets come from: each line above is a rule that can change. If the PF ceiling, a tax slab or the WPS deadline moves and the KB isn't updated, L1 quotes the old number and the ticket reopens. Slab source: ClearTax income tax slabs.

Five ideas that matter for this case

1. **Rules change on a date.** A statutory value is right or wrong *as of a date*. In real life, India raised the EPF wage ceiling from ₹15,000 to ₹25,000 a month, notified on 17 Sep 2026. A question about an August payslip and one about an October payslip have different correct answers. Our simulated data has the same kind of change, dated 1 Aug 2026.
2. **The payroll engine is the source of truth, not the help article.** The engine is updated on the effective date; a KB article is updated when someone remembers.
3. **Payroll week is a cliff.** Most tickets arrive between the 25th and the 3rd, when salaries, bank files and statutory filings are all due. Since the UAE deadline moved to the 1st of the month, a rejected WPS file on the 30th is a crisis.
4. **Two countries, two kinds of failure.** India tickets tend to be calculation disputes (PF, TDS, PT). UAE tickets tend to be payment failures (a WPS file rejected, a missing routing code). An answer that is right for one country is wrong for the other.
5. **Tickets contain personal data.** Customers paste Aadhaar, PAN, bank account, IBAN and Emirates ID numbers. Both DPDP and PDPL apply, and data can't simply be sent to any model in any country.

> **Ask the class:** "A customer writes on 20 September: 'Why is PF on my August payslip based on ₹15,000?' Is that a bug?" *Expected answer:* probably not. The ceiling changed on 17 Sep, so August used ₹15,000. To answer correctly you need the rule, its effective date and the payslip month.

Read more

- Ramco Payce product page and launch press release
- EPF wage ceiling raised to ₹25,000 (SCC Online) and history of the ceiling (Business Today)

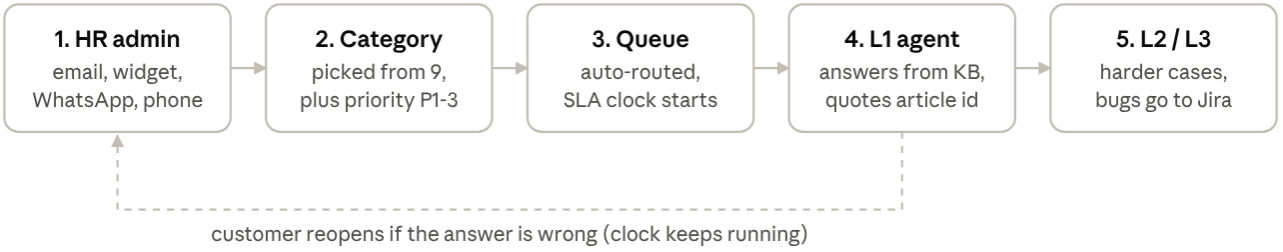
- EPFO official site: epfindia.gov.in (for ECR filing, rates and circulars)
- UAE WPS changes from 1 June 2026 (EY Global Tax News) and a complete guide to WPS (Bayut)
- UAE end-of-service gratuity (Dubai.ae)
- UAE data protection laws, including PDPL (u.ae)
- DPDP Rules 2025 explained (PIB)

Facts and links checked on Oct 23, 2026. Rules change often; check the official source before teaching a number.

How a customer raises a ticket today

The customer is usually an HR or payroll admin at a business, not the employee. They raise a ticket, pick a category and a priority, and the helpdesk sends it to a team. From that moment a contractual clock is running.

A ticket travels through five hands before it is solved



Closed when the customer confirms, or after 48 h of silence. Missed resolution target on Enterprise or Business = SLA credit.

how a ticket is raised and handled today (simulated process)

Step	What happens	Where it can go wrong
1. Raise	The admin writes in by email, the in-app widget, the partner portal (resellers), WhatsApp or phone (the agent types it up)	Five channels, five formats; phone and WhatsApp lose detail; people paste IDs and bank details
2. Categorise	The admin picks one of 9 categories (Payroll, Statutory, How-To, Reports, Leave & Attendance, Integration, Billing, Access, Onboarding) and a priority	A WPS rejection could be "Payroll" or "Statutory"; the admin guesses

Step	What happens	Where it can go wrong
3. Route	The category decides the queue automatically; the SLA clock starts	A wrong category means the wrong team
4. Answer	L1 searches the knowledge base, pastes the answer and quotes the article id	The article may describe last year's rule
5. Escalate	L2 for hard cases; L3 and Jira for product bugs	Each hand-off adds hours
6. Close or reopen	Closed on confirmation or after 48 h of silence; a reopen keeps the original clock	Reopens push tickets past their target

Tier	First response (P1 / P2 / P3)	Resolution (P1 / P2 / P3)	Credit if missed
Enterprise	1 h / 4 h / 8 h	8 h / 24 h / 72 h	10% / 5% / 2% of the monthly fee, x1.5 if more than twice the target
Business	2 h / 8 h / 24 h	12 h / 48 h / 96 h	5% / 3% / 1%
Standard	Best effort	Best effort	None

> **Ask the class:** "Which step in this flow decides whether the customer gets a credit?" *Expected answer:* all of them, because the clock runs from step 1. But steps 2 and 4 are where a person makes a choice that can quietly be wrong.

The stakeholder transcripts: everyone is telling the truth, from where they sit

The data room has nine simulated discovery calls. Nobody is lying, but each person describes the problem from their own chair, shaped by what they are measured on and what they fear. Our job is to treat every sentence as a **hypothesis** and test it against the data.

> **Say it like this:** "A stakeholder interview gives you hypotheses, not facts. The CEO's 'we need a chatbot' is a hypothesis about the solution, and so far nobody has stated the problem."

Call	Who	What they ask for	Why they say it (incentive or fear)	What to check in the data
1	Rohan Mehta, CEO	"A ChatGPT chatbot, cut support headcount 30%, before the Q4 board"	Board pressure; a competitor's launch post; thinks volume is "just the business"	Does volume explain the cost, or something else?
2	Anjali Krishnan, CFO	Lower cost per ticket; a hard cap of USD 4,500 a month on AI	Burned by a vendor pilot that cost USD 78k and spiked in peak week	Where do SLA credits actually come from?
3	Priya Nair, Head of Support	Fewer tickets, no job losses	Reports CSAT to the board and blames slow, tired agents	Is CSAT falling everywhere, or for a few customers?
4	Sandeep Rao, L2 lead	Less firefighting in payroll week	Lives in the chaos; can't measure it; careful in front of management	How often do tickets move between queues?
5	Farhan Qureshi, L1 agent	"A KB I can trust"	Pastes articles all day and gets the reopens	Do some articles lead to more reopens than others? After which date?
6	Meera Iyer, Engineering	"No write access for a bot"	Owns the payroll engine; remembers the May payslip leak	What do the APIs offer? What are their limits?
7	Karthik Subramanian, DPO	No privacy surprises	DPDP Act and PDPL liability; a payslip was emailed to the wrong customer	How much personal data sits in tickets?
8	Vikram Sethi, VP Sales	A voice bot	Lost two deals; hears "support" on sales calls	How many problems actually come by phone?
9	Two customer HR admins	Correct answers; run status without a ticket	One paid 5,100 workers late after following a help article	What did that ticket's history look like?

Three patterns to notice

1. **Everyone proposes a solution, and few describe a problem.** "Chatbot", "voice bot" and "more agents" are answers; nobody has asked a question yet.
2. **The people closest to the work say the most useful things quietly.** Farhan's "the KB says one thing, the payroll engine does another" is worth more than the board deck.
3. **The contradictions are the map.** The SOP says a ticket moves at most once; L2 says "sometimes they bounce". The runbook says the KB is updated 10 days before a rule changes; L1 says it isn't. Each contradiction is a place to dig.

Not every problem is a chatbot or RAG problem

Before choosing any AI, ask what kind of problem you have:

If the root cause is…	The fix is mostly…	AI's role
Tickets reach the wrong team	Classification and routing at intake	A small model classifying text into a fixed set of labels, with code deciding the route
Answers use outdated rules	Content ownership, effective dates, a change process	Retrieval that respects dates, and tools that ask the source system
One customer's integration is broken	An engineering bug fix	None; this is Jira, not AI
Repetitive how-to questions	Self-service and better docs	A cached, cited answer bot is fine here
Nobody notices 30 identical tickets	Incident detection	Clustering and an alert, with a human deciding
Customers want status updates	An API or webhook to show run status	A simple status page may beat any chatbot

> **Ask the class:** "If we built the CEO's chatbot on the current knowledge base tomorrow, what would happen?" *Expected answer:* it would give the same outdated answers, only faster, to more people, and with more confidence. That is what happened with the vendor pilot.

> **In plain words:** a doctor who prescribes before diagnosing is dangerous even if the medicine is good. This week we diagnose.